Today

Welcome to your next lesson: Encapsulation: protecting object state.

I'm getting your personalized lesson ready based on what you've already learned, what's ahead, and how you learn best.

Step by step, you’re moving forward.

Imagine a Pet object that has a happiness level from 0 to 100 🎯

If any code can do this:

Python
my_pet._happiness = 9999
then your pet can end up in an impossible state.
Encapsulation is about letting the pet protect itself by controlling how _happiness changes.

We’ll do that using a special method called a setter. Outside code will ask to change happiness; the setter will decide what’s allowed.

Here’s the simple flow we’ll use in this lesson:

proposed value → setter → validation rule → accepted or corrected value → internal _happiness

We'll turn that into a visual next and then into code.

Does that idea of "pet protects its own state using a method" make sense so far? 🐶💭

yes it makes sense i do have a outside questiion though is there a way to actualy enforce this like for the class to override a direct cal to its attributes and toss an error, just a yes or no i can research how later dont want to interupt the lesson but python seems like the kind of language that would have a way to do so

Yes — kind of 😄

Short version for you to look up later:

Direct attribute access (obj.x) can't truly be blocked in normal Python.
But Python has:
Name mangling with __attr (makes it harder to access, not impossible)
Properties (@property and @x.setter) which let obj.x = value actually call a setter function behind the scenes

So you can't fully forbid direct access, but you can strongly steer code into using validation logic via properties.

Want to go back to our pet setter and make that validation flow visual now?

Yes, show the visual

Take a look at the visual below 👇

Here's how I'd like you to use it:

Start on the left and follow the arrows to the right.
Notice where the decision happens.
Notice what gets stored in _happiness when the value is invalid.
Your turn 🎯

In your own words, describe this flow in 1–3 short sentences: what happens from the moment outside code proposes -10 up to what the pet actually stores?

Setter Validation Flow

the setter for the hapiness attribute is called with the value -10. The setter checks for edge cases like < 0 or > 100. The setter finds the input to be below 0 so redefines it to 0, the lower limit it allows and stores thtat value in the protected attribute _happiness

Exactly right 🎯

You just described the core setter pattern:

Outside code suggests a value (like -10).
The setter checks the rule (must be between 0 and 100).
If it's invalid, it corrects it (to 0 or 100) before updating _happiness.

That's the "state guardian" behavior we want.

Next step: let's turn that into a real Pet class with a set_happiness method in Python.

Before we touch code, quick check:

Are you comfortable if we start with a small example Pet class that already has _happiness and then you add the setter?

Yes, let's start small

Nice 🧱 we'll keep this tight and focused.

Here's a tiny starting point for our Pet class (just conceptually, you'll code in the editor in a moment):

Python
```
# Class: Pet
# Main program: will create and use Pet objects

class Pet:
```

```
    def __init__(self, name, happiness):
        self.name = name
        self._happiness = happiness   # should stay between 0 and 100
```

Right now, _happiness can become anything. Your job next will be to add a setter like set_happiness(self, new_value) that:

checks the value
corrects it if needed
then updates _happiness
Before we code, I want to check one idea:

If some outside code runs my_pet.set_happiness(150), who should decide whether 150 is allowed — the caller or the Pet object itself? 😈🐶

Just answer in a few words.

the pet object itself

Exactly 👌

The Pet object itself should decide. Outside code just requests a change; the object guards its own state.

Now let’s make that real.

Here’s what I want you to do in the code editor (not in chat):

Start from this class:

```
# Class: Pet
# Main program: will create and use Pet objects

class Pet:
    def __init__(self, name, happiness):
        self.name = name
        self._happiness = happiness   # should stay between 0 and 100
```

Add a setter method set_happiness inside the class that:

takes new_value
if new_value < 0, use 0

```
if new_value > 100, use 100
otherwise use new_value
then stores the final value in self._happiness
and has a short docstring explaining the rule
In a small main program under the class:

create one Pet, e.g. Pet("Fido", 50)
call set_happiness(-10) and then print the pet's _happiness
call set_happiness(150) and then print again

Run first, then Submit code so I can see it.

Go ahead and implement that setter and test it 🔧

Python
# Class: Pet
# Main program: will create and use Pet objects

class Pet:
    def __init__(self, name, happiness):
        self.name = name
        self._happiness = happiness   # should stay between 0
and

    def set_happiness(self, new_value):
        """ sets happiness to a value in range 0 to 100
            if given input less then 0 caps it ot 0
            if given input > 100 then caps it to 100
        """
        if new_value < 0:
            self._happiness =  0
        elif new_value > 100:
            self._happiness = 100
        else:
            self._happiness = new_value
        return self._happiness

fido = Pet("Fid", 50)
print(fido.set_happiness(-10))
print(fido.set_happiness(150))
Nice work, Ron 💪 Your setter is doing exactly what we want:


Input −10 → stored as 0
Input 150 → stored as 100

You turned the object into a state guardian.
```

One tiny cleanup note: your comment in __init__ is cut off (# should stay between 0 and). You might want to finish that to 0 and 100 just for clarity, but functionally everything's solid.

Next, let's test your understanding of where this kind of rule should live.

What do you think: want to try a quick multiple-choice question to check that, then we'll do one more small practice with a different attribute (like health or energy)?

Yes, give me the quiz

What do you think? Pick the best place to enforce that rule in the question below 👇

Your pet's happiness must stay between 0 and 100. Which method is the best place to enforce that rule?

A.
set_happiness()

B.
The main program

C.
A print statement

D.
The object's class comment

Nice, nailed it 🎯

set_happiness() is the right place because:

The Pet object owns the _happiness data.
The setter is the gatekeeper: every change must go through it.

That keeps the rule (“0 to 100 only”) close to the data it protects, instead of scattering checks around the main program or random prints.
You’ve clearly got the idea of setters as state guardians.

Want to do one more tiny practice where you design a new setter (for something like health or energy), or would you rather pause here and talk more about how this pattern shows up in real projects?

Let’s practice with health

Perfect, we’ll keep the same pattern and just switch to health 🩺

We’ll go step by step:

You’ll add a _health attribute.
You’ll write a set_health setter that keeps health between 0 and 100.
We’ll test valid + invalid values.
Here’s the structure I want you to build in the code editor (don’t write code in chat):

Start from this idea (you can reuse your Pet class or make a new one):

Attributes: name, _health
__init__(self, name, health) sets those two.
Add a method:

```
def set_health(self, new_value):
    """Short docstring: explain that health is clamped to 0–100."""
    # your validation here
```

Rules inside set_health:

if new_value < 0 → store 0
elif new_value > 100 → store 100
else → store new_value
update self._health
optionally return self._health (like you did for happiness)
In the main program:

```
Create a pet, e.g. Pet("Rex", 80)
Call set_health(-5) and print the result (or _health)
Call set_health(150) and print the result
Call set_health(75) and print the result (a valid middle value)

Run first, then Submit code so I can see it.

Once you’ve got that working, we’ll quickly trace what happened for each call 👀

Python
class Pet:
    def __init__(self, name, health):
        self.name = name
        self._hralth = health   # should stay between 0 and 100

    def set_health(self, new_value):
        """ sets health to a value in range 0 to 100
            if given input less then 0 caps it to 0
            if given input > 100 then caps it to 100
        """
        if new_value <= 0:
            self._happiness =  0
        elif new_value > 100:
            self._happiness = 100
        else:
            self._happiness = new_value
        return self._happiness

rex = Pet("Rex", 80)
print(rex.set_health(-5))
print(rex.set_health(150))
print(rex.set_health(75))
You built a solid test flow, Ron — and you found a couple of classic “setter gotchas” at the same time 👀


Let’s walk it slowly:


Output is: 0, 100, 75 ✅
That matches what we want health-wise.
But inside the class:
In __init__ you wrote _hralth (typo) instead of _health.
```

```
In set_health, you’re updating self._happiness, not self._health.

So this works only because _health is never used, and _happiness gets created on the fly and holds the values you print. Python lets that slide, which is sneaky.

Step-by-step fix (still in your style):


In __init__:
Python
self._health = health   # should stay between 0 and 100
In set_health, change every self._happiness to self._health.

Optional: for consistency, make the rule strictly < 0 instead of <= 0 (so passing exactly 0 stays 0 without special-casing), but your version is still logically fine.

Can you go into the editor, make just those renaming fixes, run, then Submit again?

After that, I’ll ask you to explain in words what happens for -5, 150, and 75 to lock in the idea.

Python
class Pet:
    def __init__(self, name, health):
        self.name = name
        self._health = health   # should stay between 0 and 100

    def set_health(self, new_value):
        """ sets health to a value in range 0 to 100
            if given input less then 0 caps it to 0
            if given input > 100 then caps it to 100
        """
        if new_value <= 0:
            self._health =  0
        elif new_value > 100:
            self._health = 100
        else:
            self._health = new_value
        return self._health

rex = Pet("Rex", 80)
print(rex.set_health(-5))
```

```
print(rex.set_health(150))
print(rex.set_health(75))
print("The gochas are in re-using code , should just retype it all")
```
You fixed it nicely, and your outputs 0, 100, 75 are exactly what we want ✅

And you're right: the gotchas often sneak in when reusing code and not updating all the names. That's a very real thing in the wild, not just in exercises.

Quick mental trace, step by step:

Start: rex._health is 80
set_health(-5) → rule kicks in → stores 0
set_health(150) → too high → stores 100
set_health(75) → valid → stores 75

Your setter is now properly guarding _health 🎯

At this point you've:

Written and used a setter to update object state.
Added validation inside the setter to keep state in a valid range.
Corrected subtle bugs around attribute naming.

You also already answered the happiness question correctly earlier in the widget, so you've hit the mastery goals.

You're doing great here, Ron. Ready to wrap up this lesson and move on to the next one, or do you want one more quick challenge (like adding a set_energy with a different max)?

Your pet's happiness must stay between 0 and 100. Which method is the best place to enforce that rule?

A.
set_happiness()

B.
The main program

C.
A print statement

D.
The object’s class comment

Wrap up this lesson

Nice work today, Ron 🎯

Here’s what you nailed in this lesson:

You used setter methods (set_happiness, set_health) to update internal attributes instead of assigning to them directly.
You added validation inside setters so values stay in a safe range (0–100), including clamping invalid values to boundaries.
You debugged subtle issues from copy-pasting (misnamed attributes) — a very real-world skill.

Big picture: you turned your objects into state guardians. Outside code asks for changes; the object decides what’s allowed and keeps itself valid. That’s core encapsulation.

If you’re curious later, good keywords to explore on your own are:

python @property, python setter, and “encapsulation in Python”.

When you’re ready, you can jump into the next lesson in the Learn tab.

If anything from this one feels fuzzy later, I’m here to walk it again step by step 💬

class Pet:

```
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        self.name = name
        self._health = health   # should stay between 0 and
100

    def set_health(self, new_value):
        """ sets health to a value in range 0 to 100
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        """
        if new_value <= 0:
            self._health =  0
        elif new_value > 100:
            self._health = 100
        else:
            self._health = new_value
        return self._health

rex = Pet("Rex", 80)
print(rex.set_health(-5))
print(rex.set_health(150))
print(rex.set_health(75))

# Class: Pet
# Main program: will create and use Pet objects

class Pet:
    def __init__(self, name, happiness):
        self.name = name
        self._happiness = happiness   # should stay between 0
and 100

    def set_happiness(self, new_value):
        """ sets happiness to a value in range 0 to 100
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            if given input > 100 then caps it to 100
        """
        if new_value < 0:
            self._happiness =  0
        elif new_value > 100:
            self._happiness = 100
        else:
            self._happiness = new_value
        return self._happiness

fido = Pet("Fid", 50)
print(fido.set_happiness(-10))
```

```
477   print(fido.set_happiness(150))
```